

Filling the Gaps?: Private Sector Response to the Push and Pull of the Indian Public Schooling System

Prerna Dewan, Subhasish Dey, Kasturi Kandalam, Abhiroop Mukhopadhyay,
Soham Sahoo

Abstract

We study how improvements in public school quality shape the evolution of local schooling markets in India. Combining district-year ASER test scores with the universe of UDISE administrative data for 2012–2017, we document a robust and economically meaningful inverted-U relationship between public school quality and private school presence. When public schools perform poorly, marginal improvements in public test scores are associated with more private school entry, a larger stock of private schools, and a rising private–public enrollment ratio. Beyond a threshold, the relationship reverses: additional gains in public school quality coincide with fewer new private schools, slower growth in the stock of private schools, and declining private enrollment shares. We investigate mechanisms by examining the quality margins along which private schools adjust. As public school quality improves, private schools that remain in the market invest in more qualified teachers and better infrastructure, both of which are costly. These convex responses are consistent with a simple mechanism; when public schools are weak, private providers can operate at low cost and high enrollment; as public schools improve, private schools face declining demand and must upgrade quality to remain viable. Taken together, our results illustrate how public-sector performance shapes the supply of private schooling in a mixed education system.

(*JEL*: I21, I25, L11, L15, M13)

Keywords: Private schools, Public schools, Evolution of market for education

1 Introduction

Private schools play an increasingly prominent role in India’s education landscape. Over the last two decades, the rapid growth of low-fee private schools has reshaped local schooling markets, expanded household choice, and generated a complex set of interactions between public and private provision. This expansion has been accompanied by a persistent debate about the relative quality of public and private schools, and its implications for efficiency and equity of schooling markets. Early work argued that observed differences in test scores largely reflect selection, with private schools enrolling students from households with more favorable inputs and backgrounds (Kingdon, 1996; Kingdon, 2007; Chudgar and Quin, 2012). A parallel strand of research, however, finds that private schools often generate learning outcomes comparable to public schools at substantially lower cost, suggesting differences in efficiency

rather than student composition (Muralidharan and Sundararaman, 2015; Singh, 2015). At the same time, concerns about the quality of public schooling have remained central to policy debates, with substantial variation across districts in both the level and trajectory of public learning outcomes. While the literature has focused primarily on comparing levels of quality across sectors, much less is known about how changes in public school quality reshape private school behavior and market structure over time. Understanding how private schools respond to improvements or declines in the public option is therefore essential for interpreting the joint evolution of schooling markets and for designing policies that improve access, efficiency, and learning.

This paper asks the question: how does the quality of public schools shape the scale, composition, and quality of the private schooling sector? While households choose between public and private schools based on price, location, and perceived quality, private schools make entry and investment decisions in response to expected demand and the cost of quality provision. Public school quality enters this equilibrium not only as a determinant of relative quality, but also as a factor that shapes household willingness to pay, the minimum quality threshold required for private schools to remain viable, and the profitability of new entry. Yet despite its centrality, the empirical relationship between public sector performance and private sector behavior remains poorly understood.

We make progress by constructing a district-year panel dataset that links the ASER measures of learning outcomes in public schools with the universe of private schools recorded in UDISE, covering the period 2012–2017. The ASER measures provide comparable, widely disseminated indicators of basic reading and math proficiency, which we treat as proxies for the effectiveness of local public schools. UDISE allows us to track the total number of private schools, new private school entry, private enrolment per school, and key quality margins such as teacher qualifications and school infrastructure. The combined dataset provides us an opportunity to study public-private interactions at a spatial-temporal scale, while controlling for district-level demographic and economic conditions.

We begin by documenting a robust and economically significant non-linearity. Across all specifications, we find an inverted-U relationship between public school quality and private school activity. In districts where public schools perform poorly, improvements in public learning are associated with more private school entry, a larger stock of private schools, and a rising private-public enrollment ratio. Beyond a threshold, however, these relationships reverse: at higher levels of public quality, further improvements coincide with fewer new private schools, slower growth in the stock of private schools, and reductions in private enrolment relative to the public sector. These patterns are stable across functional forms, alternative measures of public quality, and subsamples of primary and middle schools.

To shed light on mechanisms, we examine how private schools respond along intensive quality margins. We show that as public school quality improves, private schools invest more heavily in teacher qualifications and physical infrastructure—two margins that are both costly and salient to parents. These

adjustments are convex: at higher levels of public quality, the slope of the relationship between public quality and private investment steepens. At the same time, enrolment per private school declines, suggesting that private schools face increasing competitive pressure and shrinking demand as the public option improves. Together, these patterns are consistent with a story of vertically differentiated schooling markets: when public schools are weak, private providers can enter and operate at low cost; when public schooling becomes stronger, private schools that remain in the market must upgrade quality to maintain enrolment, while new entry becomes less attractive.

The use of lagged public test scores mitigates reverse-movement concerns; district fixed effects account for persistent differences in preferences and economic conditions; and state-year fixed effects absorb all contemporaneous state-level reforms, budget changes, and regulatory shifts. A broad set of district-year controls — including nightlight intensity, parental education, and demographic composition — further isolate the role of public school performance independent of the effect of general local development. Robustness exercises confirm that the inverted-U shape is not sensitive to functional form, subsample choice, or alternative outcome definitions.

This paper contributes to a growing body of work on public–private interactions in education and the organization of schooling markets. A large literature emphasizes the role of information frictions and parental responses to perceived school quality in shaping enrollment patterns (Andrabi, Das, and Khwaja, 2017). Complementary theoretical work models schooling markets as vertically differentiated systems in which households sort across providers and schools choose quality and scale in response to competitive pressures (Epple and Romano, 1998; Epple, Romano, and Sieg, 2006). Related empirical research highlights the importance of local market structure and spatial competition—showing that school behavior depends not only on public–private distinctions but also on the density, proximity, and quality distribution of nearby schools (Hoxby, 2000; Neilson, 2013). In contrast to much of this work, which focuses primarily on household choice or equilibrium allocations across schools, we examine how private schools adjust their entry decisions, scale, and quality investments in response to changes in the quality of the public option. By jointly studying extensive margins (private school entry and the stock of private schools) and intensive margins (enrolment per school, teacher qualifications, and infrastructure) we provide new evidence on how private providers reposition themselves as local competitive conditions evolve. Our results underscore the importance of viewing schooling markets as equilibrium systems in which improvements in public provision reshape private incentives not only through direct public–private competition, but also through changes in local market structure and competitive pressure among private schools themselves.

The remainder of the paper proceeds as follows. Section 2 provides background on India’s schooling system and descriptive trends in public and private sectors. Section 3 describes the data sources and variable construction. Section 4 outlines the empirical strategy. Section 5 presents the main results on private school entry and scale. Section 6 examines intensive-margin responses and discusses mechanism. Section 7 reports robustness analyses. Section 8 concludes with implications for policy and schooling market design.

2 Background

The Indian Education system is vast and reflects persistent tension between the government’s mandate of universal education, with the *Sarva Shiksha Abhiyaan* programme, and rapid concurrent sprouting of private schools. The relationship between public school quality and private schools proliferating can be understood as a co-evolutionary process. This paper contributes to this understanding by empirically testing how changes in public school quality affect private school entry across districts.

Early work by [Kingdon \(2007\)](#) and [Chudgar \(2009\)](#) established that the growth of private schools in India is not confined to urban or affluent populations. Even in rural areas, low-fee private schools have emerged as viable alternatives to the public option, driven by parental dissatisfaction with public school deficiencies such as teacher absenteeism, multi-grade classrooms, and limited accountability.

[Pal \(2010\)](#) shows that private schools are more likely to locate in villages with better local public infrastructure, which includes, among other things, access to piped water, electricity, phones and a post office. This is shown to be driven in part by greater demand from households with greater education and from those belonging to higher castes, and these are unequally distributed across villages. [Pal \(2010\)](#) also documents higher average pass rates in villages with better public infrastructure and private school presence, but notes an insignificant impact of greater private school presence on the pass rates of existing public schools in these villages. More recent work like [Bagde et al. \(2022\)](#) also suggests that private expansion has led to little to no improvement in public outcomes.

The “push” and “pull” of public education demonstrate not just the inner workings of the market, but social negotiations between the state and private players. Debates on school quality in India often hinge on whether private provision outperforms public schools once we adjust for student background and sorting. [Kingdon \(1996\)](#) finds, in urban India, that raw score comparisons are misleading as they do not account for endogenous selection, and estimates achievement functions that correct for selection and home inputs before attributing differences in quality or efficiency to management type. This exercise shows why private schools may sometimes look better in simple comparisons yet converge toward parity when selection is controlled for. More recently, [Central Square Foundation \(2020\)](#), in a state of the sector report, underscores the sheer scale of private enrollment and emphasizes that learning advantages are not consistently higher after accounting for student background; it argues for outcome-focused regulation rather than input compliance.

With the rapid expansion of the private sector in school education, including the proliferation of low-fee private schools, parents have been choosing in relatively low-information markets. Field evidence from four states ([Azim Premji University Research Group, 2018](#)) shows that families greatly value character-

istics of schools such as English-medium, perceived discipline, and safety; yet perceptions often diverge from reality (e.g., “English-medium” may be in name only, or less-qualified teachers in schools chosen for “teacher quality”). [Afridi et al. \(2020\)](#) conduct an informational intervention which shows that there is a re-sorting of children across schools when parents and schools randomly receive information about the test scores of all students in a local market. This intervention results in strong statistically significant improvements in average test scores, showing that school choice decisions are made with incomplete information.

Prior work on market dynamics shows that rural entry of private high schools induces sharp stratification by achievement, caste, gender, and affluence; nevertheless, large-scale administrative data show no short-to-medium-run declines for public-school students left behind, suggesting competitive pressures and resource reallocation can be offset by peer-group losses ([Bagde et al., 2022](#)). Upstream supply shocks matter too: the rollout of elite public colleges appears to catalyze local private-school entry and shift students from public to private institutions, likely via improved infrastructure and reduced travel/setup costs ([Jagnani and Khanna, 2020](#)).

Demand mechanics are central. In nearby Pakistan’s dense low-fee private markets, families place a high premium on proximity (strong distance disutility), face relatively low own-price elasticities, and derive sizable consumer surplus from having a private option at all ([Carneiro et al., 2022](#)). Inside households, interpreting expenditure “parity” requires care: gender bias can operate through the extensive margin (non-enrollment for girls) even where per-enrolled-child spending looks equal, so detection methods matter ([Kingdon, 2005](#)).

3 Data

Our empirical analysis draws on multiple administrative and geospatial datasets spanning 2011–2017. The primary source is the District Information System for Education (DISE), the annual census of all registered schools in India. DISE provides detailed information on schools—including management type, category, year of establishment, and infrastructure (such as piped water, sanitation facilities, electricity, libraries, computers, boundary walls, and playgrounds). It also includes data on student enrollment and teacher characteristics (number, qualifications, etc.). Using the School ID variable and the year of establishment, we construct a school-level panel dataset.

A complication arises because School IDs are not always stable over time: the same school may appear under multiple codes across years, or conversely, the same code may correspond to different schools. Furthermore, when a geographical unit (such as a state, district, block, or village) is bifurcated, both the parent and child units can appear, potentially leading to double-counting. To address this, we systematically eliminate both the parent and child districts in years of bifurcation, ensuring that totals are calculated without duplication at the district–year level.

Starting in 2012, DISE integrated primary and middle school data with secondary school data and upgraded its data collection infrastructure, creating the Unified District Information System for Education (UDISE). Since then, self-reporting by private schools has improved steadily, and by 2017, UDISE achieved full coverage of the school universe ([Ministry of Education, 2022](#)). Leveraging this coverage, we construct procedures that yields two versions of a temporally consistent school-year panel for private schools. In both constructions, schools are identified by their reported school codes and expanded into school-year observations based on establishment information.

In the first construction (the union panel), a private school is treated as active in year t if its reported establishment year is at most t . Once a school appears in a year, it is carried forward through 2017. This construction gives us an upper bound on the stock of private schools by abstracting from exit mechanisms and is useful for assessing the role of entry and identifier instability. In the second construction (the panel with exits), we additionally restrict a school’s active years to those up to its last observed year in UDISE. Permanent disappearance from the administrative records is therefore interpreted as market exit. This yields a comprehensive, non-repeating panel in which each school is observed from its year of establishment (or first appearance) until its final recorded year.

We aggregate to the district–year level to compute our main dependent variable: the total number of private schools. Unless otherwise stated, all main results use the panel with exits. Figure 1 contrasts raw and cleaned national private-school counts over time. Raw administrative counts increase sharply, reflecting mechanical inflation from identifier changes and bifurcations. In contrast, the cleaned series exhibits substantially lower volatility and evolves gradually, indicating that the cleaning procedure restores temporal coherence without imposing smoothness by construction. In the cleaned data, fluctuations are evident—most notably a sharp dip in 2015 followed by a rebound in 2016—the overall trajectory is downward. Between 2012 and 2017, the total fell from roughly 2,84,425 to 2,75,956 schools, a net loss of about 8,469 private schools (a decline of approximately 3%). The cleaned series should not be interpreted as the true level of private schools, but as a temporally consistent measure that removes mechanical inflation over time.

Figure 2 shows the distribution of district-level year-to-year changes in private-school counts. Raw administrative data exhibit substantially greater dispersion and a positive shift in the distribution, consistent with mechanical inflation from identifier changes and bifurcations. In contrast, the cleaned data display a much tighter distribution centered near zero, indicating improved temporal coherence without eliminating genuine entry and exit. Outside values are excluded from the figure for visual clarity.

We matched the annual census of Indian schools, the UDISE, 2012-2017 to the Population Census (2011) for village-level amenities data. We also match the Visible Infrared Imaging Radiometer Suite (VIIRS)

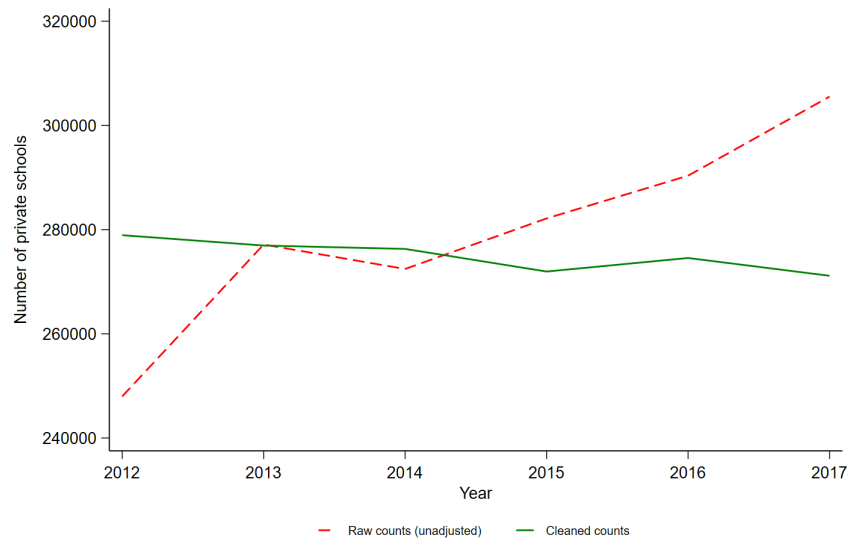


Figure 1: Total no of private schools over the years

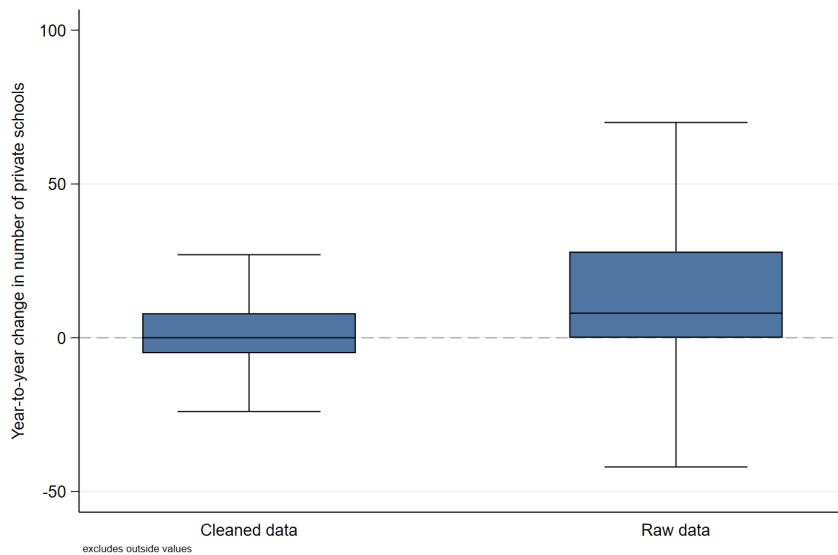


Figure 2: Total no of private schools over the years

nighttime lights data, which was processed to create annual district-level measurements.¹ Spatial data from the Census and the World Bank were also matched. The ASER dataset (2011–2017) was merged with the remaining datasets for district-level educational indicators, such as reading and mathematics learning outcomes. As ASER is conducted only in rural areas, the analysis sample is correspondingly restricted to rural districts. ASER was not carried out in 2015 and 2017. So, we include 2011 data as well to be able to use lagged scores for data in 2012.

The resulting panel dataset covers 538 districts over the 2011–2017 period. Table 1 summarizes sample attrition across data construction stages. The excluded column reports the number of districts dropped at each stage relative to the immediately preceding sample, rather than cumulative exclusions. The final dataset contains harmonized information on educational infrastructure, learning outcomes, economic activity, and spatial characteristics, and is well-suited for analyzing district-level educational dynamics.

Table 1: District Sample Size by Data Construction Stage

Dataset Construction Stage	No. of Districts	Excluded at Stage
DISE (2012–17)	693	–
Removal of bifurcated districts (parent and child)	570	123
Collapsing to district-year level	569	1
Merging with VIIRS Night Lights data (2012–17)	569	0
Merging with Admin, Social Background and Connectivity Census data (2011)	569	0
Merging with Village amenities Census data (2011)	561	8
Merging with ASER data (2011–16)	538	23

¹Monthly sub-district-level data was aggregated and harmonized to align with Census 2011 district boundaries.

4 Private Schools

This section provides descriptive evidence on the size, composition, and variation of schooling markets across districts in India. We document key sources of variation, and motivate the empirical specifications that follow. Throughout, we emphasize heterogeneity across districts rather than national averages, reflecting the localized nature of schooling markets.

4.1 Size and Composition of Schooling Markets

We consider the relative size of public and private schooling sectors across districts first. Public schools account for the majority of total enrollment in nearly all districts, but private schools constitute a substantial and highly heterogeneous share of local schooling markets. In the median district-year, private schools represent approximately percent of all schools, with the interquartile range spanning from to percent. In some districts, private schools are nearly absent, while in others they form a sizeable fraction of total provision.

Figure plots the distribution of the number of public and private schools across districts. While public schools dominate in absolute numbers, the dispersion in private school counts is considerably larger. This dispersion persists even after normalizing by district population or total enrollment, suggesting that variation in private school presence reflects more than differences in district size alone. Instead, it points to differences in local demand conditions, institutional environments, and the perceived quality of public provision.

Private school presence also varies systematically across states, but substantial heterogeneity remains within states. Districts in the same state—and subject to the same education policies and regulatory frameworks—exhibit markedly different private school shares. This within-state variation motivates our empirical strategy, which exploits differences in public school quality across districts within the same state and year.

4.2 Variation in Public School Quality

Public school quality, measured using district-level averages of ASER reading and mathematics scores, exhibits substantial variation across both space and time. Figure plots the distribution of lagged public school ASER scores across districts. While average learning levels remain low in absolute terms, the distribution is wide, with meaningful differences between districts at the lower and upper tails.

Importantly, public school quality is not static. Over the period 2012–2017, many districts experience nontrivial changes in average ASER scores, though the magnitude and direction of these changes vary considerably. Figure illustrates this heterogeneity by plotting district-level changes in public school ASER scores over time. Some districts exhibit steady improvements, others stagnate, and a smaller subset experiences declines.

These changes in public school quality are not perfectly correlated with observable indicators of economic development. For example, while districts with higher nightlight intensity tend to have higher average ASER scores, the correlation is far from one-to-one. Conditional on state fixed effects, substantial variation in public school learning outcomes remains unexplained by standard proxies for income or demographic composition. This residual variation forms the basis of our empirical analysis.

Taken together, these patterns indicate that districts differ meaningfully in both the level and trajectory of public school quality. This variation provides a natural setting for examining how private schools respond to changes in the quality of the public option within localized schooling markets.

5 Empirical Model

$$Y_{dt} = \alpha_0 + \alpha_1 \cdot T_{dt-1}^g + \alpha_2 \cdot T_{dt-1}^{g^2} + \alpha_3 \cdot \mathbf{W}_{dt-1} + \alpha_4 \cdot X_{dt} + \alpha_5 \cdot \mathbf{Z}_d + \alpha_6 \cdot \mathbf{Z}_d \cdot t + \gamma_{1_s} + \gamma_{2_t} + \gamma_{3_{st}} + \epsilon_{dt} \quad (1)$$

$$E_{dt} = \beta_0 + \beta_1 \cdot T_{dt-1}^g + \beta_2 \cdot T_{dt-1}^{g^2} + \beta_3 \cdot \mathbf{W}_{dt-1} + \beta_4 \cdot X_{dt} + \beta_5 \cdot \mathbf{Z}_d + \beta_6 \cdot \mathbf{Z}_d \cdot t + \delta_{1_s} + \delta_{2_t} + \delta_{3_{st}} + \varepsilon_{dt} \quad (2)$$

The main outcome variables of interest that we examine are: (1) Y_{dt} , the number of private schools in a district d in year t , and (2) E_{dt} , the number of new private schools in district d at time t , i.e., entry in district d in year t . T_{dt-1}^g is the average combined reading and math ASER score for public school students of district d in year $t-1$. In specification (1), $\alpha_1 + 2 \cdot \alpha_2 \cdot T_{dt-1}^g$ is the marginal effect of an increase in average public school students' ASER scores on the number of private schools. Similarly, in specification (2), $\beta_1 + 2 \cdot \beta_2 \cdot T_{dt-1}^g$ is the marginal effect of an increase in average public school students' ASER scores on the number of new private schools. The marginal effects are themselves functions of the average public school students' ASER scores due to the quadratic specification.

$\mathbf{W}_{dt-1}$ is a vector of lagged controls at the district-year level, which includes variables such as average parental education, total number of public schools, average total enrollment in public schools, average infrastructure index² in public schools, and average night light intensity. These variables can presumably affect both ASER score averages of public school students (either through the education production function or through income effects) and the number of private schools in a district by their effect on demand for private schooling. X_{dt} is the number of blocks and villages in district d in year t ; it is added as a control for the size of the district, and also to account for the measurement error introduced by double counting of schools due to bifurcation of blocks and villages³. $\mathbf{Z}_d$ is a vector of time-invariant district-level controls such as the value of an infrastructure index⁴ for district d in the census year 2011,

²Calculated using principal components analysis of amenities available in each school as reported in DISE. These amenities include the presence of drinking water, boundary walls, computers, functioning electricity, ramps, classrooms in good condition, toilets for boys and girls, computer-aided learning, medical checkups and meals in schools.

³When there is an administrative bifurcation of blocks or villages, there is a change in the unique identifier (school code) of schools in DISE. As a result, two different school codes may represent the same school in different years due to a change in the administrative division it is part of.

⁴Calculated using Census variables on connectivity, health and sanitation.

the percentage of population of district d that is rural in 2011, the percentage of population of district d that is SC in 2011 and the percentage of population of district d that is ST in 2011. Trends of these variables are also included in the specification.

γ_{1_s} and δ_{1_s} represent state fixed effects, γ_{2_t} and δ_{2_t} represent year fixed effects, and $\gamma_{3_{st}}$ and $\delta_{3_{st}}$ represent state-year fixed effects in the respective equations. These fixed effects partial out the effect of confounders that vary at both the state and year level, such as state education policy. In the appendix, we display the results of the main specification with more general district fixed effects which control for all possible district-level confounders. We find that district fixed effects explain a high proportion of the variation in the dependent variables, with an R^2 value of 0.99 when district fixed effects are included. Since the panel spans only six years of data, temporal variation within districts is limited. Including district fixed effects thus leaves very little variation for identifying the marginal effect of average public school students' ASER scores on private school numbers and entry. Thus our main specification uses district-year level excess variation in average public ASER scores above state-year averages, in addition to controlling for several district-level variables (both time-invariant $\mathbf{Z}_d$ and time-variant $\mathbf{W}_{dt-1}$) to achieve identification.

The use of lagged values of average ASER scores instead of contemporaneous values is done in order to avoid feedback effects. It is well-documented that test scores of students in private schools are higher than of those in public schools, primarily through selection (add references). As a result, having a greater number of private schools and greater enrollment in private schools can presumably affect district-wide average public school scores. The effect may be negative or positive depending on whether the selection of more able students in private schools leads to worse peer effects for public school students, or if competition from private schools and greater number of resources per child in public schools boosts test scores for public school students. Using past values of district-wise ASER average score ensures that the correlation observed is not driven by reverse causality. Moreover, in a recently published paper, [Bagde et al. \(2022\)](#) show that the entry of private secondary schools into 240 Indian villages (which were previously served by public schools only) has no effect on the scores of public school students. Similarly, [Michaud-Leclerc \(2022\)](#) also finds, in the context of rural Pakistan, no negative impact of private school expansion on test scores of public school students in grades 3-5, and no negative impact in the long term either. These results dispel any remaining concerns about the possibility of reverse causality affecting our estimates.

6 Results

6.1 Number of Private Schools (Stock and Entry)

Table 3 presents our main results, with the 'total number of private schools in district d at time t ' and the 'number of new private schools in district d at time t (entry)' as outcome variables. Column 2 shows an insignificant effect of public school quality on the number of private schools - the linear specification, however, masks heterogeneity. The third column, with a quadratic specification, shows that there exists

a statistically significant inverted U-shaped relationship between public school quality and the number of private schools. At low levels of public school quality (measured by the lagged average ASER score of students in public schools), there is a positive correlation between public school quality and the number of private schools. At higher levels of public school quality, the relationship reverses. In other words, in districts where the public option for schooling produces low average test scores, more private schools compete as the quality of public schools increases. In districts with high average test scores for public schools, incremental increases in test scores are associated with fewer private schools in the market. The marginal effects and predicted values of outcome variable for different values of lagged average ASER scores corresponding to the specification in column 3 are plotted in Figure 3. These patterns reveal that the private schooling market is competitive in districts where public schooling is wanting, and are fewer in number when the existing public option is strong.

Column 6 of Table 3 suggests that there is a similar pattern in the relationship between public school quality and the number of *new* private schools in a year, i.e., entry. Figure 4 plots the marginal effects of the lagged average ASER scores of public schools on the entry of new private schools and the predicted entry of new private schools at different levels of the lagged average ASER scores of public schools. The similar patterns in Figures 3 and 4 suggest that the effect on total number of private schools may be driven by entry. [Enter reference about Kala Krishna’s latest paper about regulatory environment making exit of firms difficult – implying exits may not respond to public school quality as flexibly as entry of new private schools might.]

Taken together, these results suggest that changes in the number of private schools, and especially in new private school entry, are responsive to the quality of public education. We next examine whether the relative scale of private schooling—measured by the ratio of enrollment in private schools to that in public schools—shows a similar pattern. This provides a sense of how household demand shifts across school types in response to public school performance.

6.2 Ratio of enrolment in private schools to enrolment in public schools

Table 7, Table 8 and Figures 8 and 9 present these results. The outcome variable is the ratio of normalized enrollment in private schools (and in new private schools for the entry specification) to normalized enrollment in public schools in district d at time t . The second panel of Figure 9 plots the predicted values of this ratio against lagged average ASER scores of public school students, while the first panel (Figure 9) shows the corresponding marginal effects.

The point estimates trace an inverted-U pattern similar to that observed for the number of private schools and for entry, though the confidence intervals are wider and encompass zero over much of the range. This statistical imprecision likely reflects the short time span of our data and the fact that aggregate enrollment adjusts more slowly than the extensive margin of entry. If school fees and switching costs for households are sticky within a few years, when private profitability conditions change, student reallocation across

sectors may be slow to adjust. Despite the broad confidence bands, the central estimates are economically consistent with the earlier findings: at low levels of public school quality, improvements in public test scores are associated with a rising ratio of private to public enrollment, while at higher levels of public quality, the slope turns negative. This suggests that private enrollment expands where public schooling is weak but contracts once the public option becomes sufficiently effective. The results, while imprecisely estimated, reinforce the interpretation that the private sector’s footprint in schooling markets reflects the relative profitability of operating vis-à-vis public schools.

7 Robustness

7.1 Different School Categories

Table 4 presents results of the main specification for the subsample of the primary schooling market, i.e., it regresses the total number of private primary schools on the lagged average ASER scores in public primary schools while accounting for a range of district- and district-year- level controls, and year, state, and state-year fixed effects. The quadratic specification reveals a similar pattern as in the case of all schools; the number of private primary schools increases and then decreases as lagged average ASER scores in public primary schools increase. Below a certain threshold of public primary school quality, private primary schools compete with existing public options whereas the number of private primary schools declines as public primary school quality increases beyond the threshold. Figure 5 plots the marginal effect of public primary school quality and the predicted number of private primary schools for different levels of lagged average ASER scores in public primary schools. Confidence intervals are wider for lower levels of public school quality.

Similarly, Table 5 reveals the same association between the quality of existing public middle schools and the total number of private middle school; their relationship also exhibits an inverted U-shape.

8 Cost Margins and Private Investments

8.1 Quality of Private Schools

Table 10 presents results from our main specification where the dependent variable is the average ASER score of students enrolled in private schools in district d at time t . The key explanatory variable is the lagged average ASER score of students enrolled in public schools, along with its quadratic term, and we continue to include district-level and district-year-level controls, as well as year, state, and state-year fixed effects. In contrast to earlier results on private school quantity and entry, we find that private school quality exhibits a monotonically increasing—and statistically significant—relationship with public school quality. Both the marginal effects and the predicted values plotted in Figure 11 show that as public school quality rises, private school learning outcomes also increase, and at an increasing rate.

This pattern suggests that improvements in public schooling may raise the overall equilibrium quality of education in the local market, possibly by shifting parental expectations, intensifying reputational competition, or increasing the opportunity cost of low-quality provision among private schools. However, the convex shape also points to rising costs of maintaining and improving quality: as public school performance improves, only those private schools capable of incurring higher costs—through better teachers, infrastructure, or pedagogical investments—are likely to sustain or expand. The result is consistent with a setting where quality improvements for private schools are costly, and where competition from an improving public system raises the quality threshold needed for survival and differentiation.

8.2 Teacher Quality

Table 9 presents the results when the outcome variable is the proportion of teachers in private schools with at least a graduate degree in district d at time t . The key independent variable is the lagged average ASER score of students enrolled in public schools, entered with both linear and quadratic terms, alongside the standard set of district-level and district-year-level controls, and year, state, and state-year fixed effects. The estimates reveal a strong, statistically significant, and upward-sloping relationship between public school quality and the educational qualifications of teachers employed in private schools. Both the marginal effects and the predicted values plotted in Figure 10 show that as public school quality improves, the share of graduate teachers in private schools rises—and at an increasing rate.

This convex relationship implies that as the public schooling system becomes more effective, private schools respond by upgrading the quality of their teaching workforce, possibly as a means of differentiation or to justify higher fees. The pattern is consistent with the notion that quality improvements in private schools are costly: employing better-qualified teachers increases salary and recruitment expenses, so such investments become viable only when competition from improved public schooling compels private schools to match or exceed those standards. The accelerating slope suggests that at higher levels of public school quality, maintaining competitiveness requires disproportionately greater investments in teacher quality, reflecting rising marginal costs of quality enhancement within the private sector.

8.3 Infrastructure Index

Table 6 presents the results where the dependent variable is the private school infrastructure index in district d at time t , constructed as a standardized composite (using Principal Components Analysis) of key infrastructure indicators such as availability of classrooms, toilets, electricity, boundary walls, computers etc. The main explanatory variable remains the lagged average ASER score of students enrolled in public schools, included in both linear and quadratic form, and the regression continues to control for the full set of district-level and district-year-level covariates, as well as year, state, and state-year fixed effects. The estimates indicate a positive and statistically significant relationship between public school quality and the quality of private school infrastructure. Both the marginal effects and predicted values shown in Figure 7 rise monotonically, and the latter rises at an increasing rate, as lagged public school quality improves.

This convex pattern suggests that improvements in public schooling are accompanied by accelerated investment in physical infrastructure within the private sector. As public schools become stronger competitors, private providers appear to respond by upgrading the physical environment of their schools—perhaps to sustain their reputational advantage, attract higher-paying students, or signal quality credibly in a market where learning outcomes are only imperfectly observable. The increasing slope of the relationship is consistent with the idea that such infrastructural upgrades are costly and that the marginal cost of competing on quality rises as public schools improve. Together with the results on teacher quality, these findings underscore that the private sector’s response to a more capable public schooling system operates through cost-intensive quality enhancement rather than through price or quantity adjustments alone.

9 Discussion and Conclusion

The empirical patterns documented above point to a systematic, non-linear response of private schools to changes in public school quality. While the reduced-form estimates do not identify a single causal channel, the joint behavior of extensive- and intensive-margin outcomes helps discipline the set of plausible mechanisms.

A first class of explanations emphasizes profitability and competitive pressure in vertically differentiated schooling markets. When public schools perform poorly, even modest improvements in public learning outcomes may increase overall demand for schooling without substantially improving the outside option faced by private providers. In this regime, private schools can attract students at relatively low cost, encouraging entry and expansion. As public school quality continues to improve, however, the public option becomes a closer substitute for low- and mid-quality private schools. Private enrolment per school declines, entry slows, and the marginal private school faces increasing pressure to differentiate on quality. The observed convex increases in teacher qualifications and infrastructure investments are consistent with this mechanism, reflecting costly quality upgrades undertaken by private schools that remain in the market.

A second possibility is a market-composition mechanism, whereby improvements in public school quality selectively displace low-quality or informal private providers. In this view, the decline in private entry and scale at higher public quality levels reflects exit or non-entry of marginal private schools, while the remaining private schools are higher-quality by construction. Two features of the results, however, weigh against pure composition effects. First, quality-related outcomes rise smoothly rather than discontinuously with public school quality, suggesting incremental adjustment rather than discrete turnover. Second, similar patterns emerge within primary and middle school subsamples, which differ in baseline private-school composition.

A third explanation centers on household sorting and demand reallocation. As public schools improve,

households—particularly those at the margin between low-fee private schools and public schools—may re-enroll in the public sector. Such sorting would reduce private enrolment and entry while increasing the average quality demanded by remaining private-school households, inducing higher investment in teachers and infrastructure. While sorting likely plays a role, this mechanism alone cannot account for the sharp decline in enrolment per private school observed at higher levels of public quality, which points to shrinking market size rather than merely changing clientele.

Taken together, the combination of declining private enrolment per school, reduced entry at higher public quality levels, and convex upgrading along costly quality margins is most consistent with a profitability-based mechanism in which improvements in the public option compress private margins and raise the minimum quality threshold required for private schools to remain viable. While alternative mechanisms may operate simultaneously, the internal coherence of responses across margins suggests that changes in public school quality fundamentally reshape private incentives rather than simply reallocating students across sectors.

These findings have several implications for how public investments in education interact with private provision. First, they suggest that improvements in public school quality need not generate a monotonic crowding-out of private schools. Instead, public quality improvements can initially coexist with, and even stimulate, private entry and expansion when public schools are sufficiently weak. In this range, private schools appear to operate as close but imperfect substitutes, serving unmet demand and responding to rising concern about learning outcomes. Only once public schools reach a higher level of effectiveness does the competitive pressure exerted by the public option begin to compress private margins and discourage entry.

Second, the results underscore the importance of distinguishing between extensive and intensive margins when evaluating the impact of public-sector reforms. While the number of private schools and private enrolment decline at higher levels of public quality, private schools that remain in the market respond by upgrading teacher qualifications and physical infrastructure. These adjustments suggest that competition from improved public schools shifts the private sector toward higher-cost, higher-quality provision rather than simply inducing exit. From a welfare perspective, this implies that the consequences of public school improvements for private schooling cannot be assessed solely through changes in the number of private schools or enrolment shares, but must also account for changes in the quality of private provision.

Third, the non-linear nature of the response highlights the relevance of local market structure in shaping equilibrium outcomes. Districts with low public school quality appear to support a relatively large number of low-cost private providers, whereas districts with stronger public schools sustain fewer private schools that operate at higher quality and cost. This pattern is consistent with schooling markets characterized by vertical differentiation and heterogeneous providers, and suggests that public quality improvements can reallocate private supply across segments rather than eliminating private provision altogether.

Several limitations of the analysis warrant discussion. The relatively short time span of the panel limits our ability to observe long-run adjustments, particularly along slow-moving margins such as enrolment and school exit. Data constraints also prevent us from observing private school fees, closures, or within-school changes in inputs over time, which would allow for sharper discrimination between within-school upgrading and compositional change. In addition, while lagged public school test scores and rich fixed effects mitigate concerns about reverse causality and confounding trends, the estimates should be interpreted as descriptive equilibrium relationships rather than causal effects of specific policy interventions. These limitations point to promising directions for future research. Linking administrative school data with information on fees, household beliefs, and regulatory enforcement would allow for more direct tests of the mechanisms emphasized here. School-level panel data would further enable modeling of entry and exit decisions and the dynamics of quality upgrading. Finally, exploiting quasi-experimental variation in public school quality — such as policy-driven changes in teacher deployment, or school inputs, or public school rationalisation — would provide sharper causal estimates of how private providers respond to public-sector improvements.

In conclusion, this paper provides new evidence on how changes in public school quality shape the structure and behavior of private schooling markets in India. Rather than inducing uniform crowding-out, improvements in public schooling generate a non-linear set of responses that operate through both entry decisions and costly quality adjustments. These findings highlight the importance of viewing public and private schools as part of a single, interdependent system, and suggest that policies aimed at improving public school quality can have nuanced and heterogeneous effects on private provision. Designing effective education policy therefore requires accounting not only for direct effects on public schools, but also for the equilibrium responses of private actors operating alongside them.

References

- Afridi, F., Barooah, B., and Somanathan, R. (2020). Improving learning outcomes through information provision: Experimental evidence from indian villages. *Journal of Development Economics*, 146:102276.
- Azim Premji University Research Group (2018). School choice in low-information environments: A study of perceptions and realities in four states.
- Bagde, S., Eppe, D., and Taylor, L. (2022). The emergence of private high schools in india: The impact of public-private competition on public school students. *Journal of Public Economics*, 215:104749.
- Carneiro, P., Das, J., and Reis, H. (2022). The value of private schools: Evidence from pakistan. Technical Report 22/091, RISE Working Paper Series.
- Central Square Foundation (2020). State of the sector report on private schools in india.
- Chudgar, A. (2009). Private schooling and achievement for children in india: Possibilities and limits. *Comparative Education Review*, 53(2):155–177.

- Jagnani, M. and Khanna, G. (2020). The effects of elite public colleges on primary and secondary schooling markets in india. *Journal of Development Economics*, 146:102512.
- Kingdon, G. G. (1996). The quality and efficiency of private and public education: A case-study of urban india. *Oxford Bulletin of Economics and Statistics*, 58(1):57–82.
- Kingdon, G. G. (2005). Where has all the bias gone? detecting gender bias in the intrahousehold allocation of educational expenditure. *Economic Development and Cultural Change*, 53(2):409–451.
- Kingdon, G. G. (2007). The progress of school education in india. *Oxford Review of Economic Policy*, 23(2):168–195.
- Michaud-Leclerc, C. (2022). Private school entry, sorting, and the performance of public schools: Evidence from pakistan. Unpublished manuscript.
- Ministry of Education (2022). UDISE+ 2021–22: Flash Statistics. Technical report, Department of School Education & Literacy, Government of India. Unified District Information System for Education Plus.
- Pal, S. (2010). Public infrastructure, location of private schools and literacy outcomes: Evidence from india. *Review of Development Economics*, 14(1):24–44.

Figures and Tables

Table 2: Summary Statistics

	Mean	Median	Standard Deviation	Minimum	Maximum
Total Private Schools	508.471	305.500	588.943	1.000	7001.000
New Private Schools (Entry)	17.962	8.000	26.265	0.000	281.000
Private School Enrolment	8.67×10^{36}	1.67×10^{33}	2.83×10^{37}	0.000	1.58×10^{38}
Public School Enrolment	3.21×10^{36}	1.75×10^{32}	1.61×10^{37}	1.69×10^{19}	1.51×10^{38}
Private School Infrastructure Index	3.114	3.081	0.877	1.000	5.000
Graduate Teacher Share in Private Schools	0.019	0.016	0.010	0.000	0.061
Public Schools' Average ASER Score	3.428	3.433	0.441	1.825	4.525
Private Schools' Average ASER Score	3.748	3.743	0.327	1.000	5.000
Nightlight Intensity	0.000	-0.155	1.000	-0.164	22.493
Average Parental Education	9.594	9.611	1.036	5.701	12.959
Share of SC population	0.006	0.004	0.008	0.000	0.075
Share of ST population	0.040	0.001	0.214	0.000	4.212

Table 3: Effect of lagged ASER learning levels in public schools on total number of private schools and entry of new private schools

	Total No. of Private Schools			Private School Entry		
	(1)	(2)	(3)	(4)	(5)	(6)
Avg ASER learning in public schools _{t-1}	-281.3*** (51.60)	-54.57 (45.70)	1224.1*** (399.1)	-11.41*** (1.812)	-6.304*** (2.162)	63.17*** (14.63)
Avg ASER learning in all schools _{t-1} ²			-192.7*** (57.18)			-10.47*** (2.165)
Average Parental Education _{t-1}		50.37** (19.86)	46.26** (19.21)		0.529 (0.951)	0.306 (0.918)
Total no. of public schools _{t-1}		-0.0220 (0.0344)	-0.0171 (0.0337)		0.00701*** (0.00210)	0.00728*** (0.00204)
Normalized enrollment in public schools _{t-1}		-5.563 (10.21)	-5.463 (10.06)		-0.0665 (0.736)	-0.0610 (0.721)
Infrastructure index in public schools _{t-1}		30.25 (44.35)	27.40 (43.37)		-0.395 (1.893)	-0.549 (1.832)
Light Intensity _{t-1}		319.7** (129.0)	303.8** (129.8)		8.760*** (2.565)	7.891*** (2.535)
Census infra index		-154.8*** (48.22)	-161.3*** (49.60)		-6.681** (3.198)	-7.034** (3.225)
Census infra index $\times$ Trend		14.31*** (5.382)	14.38*** (5.516)		0.795 (0.517)	0.799 (0.540)
Log(Rural Population)		375.9*** (72.77)	379.6*** (72.11)		17.23*** (4.095)	17.43*** (4.055)
Log(Rural Population) $\times$ Trend		-11.72 (9.869)	-13.19 (9.976)		-1.429** (0.604)	-1.509** (0.608)
% of SC population		11783.4*** (3545.2)	11792.3*** (3585.2)		524.9** (207.9)	525.4** (207.7)
% of SC population $\times$ Trend		-669.3 (525.3)	-720.5 (548.6)		-34.99 (31.34)	-37.77 (32.46)
% of ST population		204.9*** (66.48)	189.2*** (65.16)		12.74*** (4.072)	11.89*** (4.044)
% of ST population $\times$ Trend		30.96 (22.12)	33.23 (22.58)		-0.588 (0.875)	-0.465 (0.899)
Constant	1492.4*** (184.3)	-2353.9*** (549.5)	-4347.7*** (915.5)	57.05*** (6.351)	-47.35** (21.73)	-155.7*** (34.42)
Observations	2514	1615	1615	2019	1615	1615
R-squared	0.0426	0.753	0.760	0.0394	0.500	0.510
Mean of Dependent Variable	528.6	501.8	501.8	18.30	17.39	17.39
# of Blocks & Villages	NO	YES	YES	NO	YES	YES
Year FE	NO	YES	YES	NO	YES	YES
State FE	NO	YES	YES	NO	YES	YES
State FE $\times$ Year FE	NO	YES	YES	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

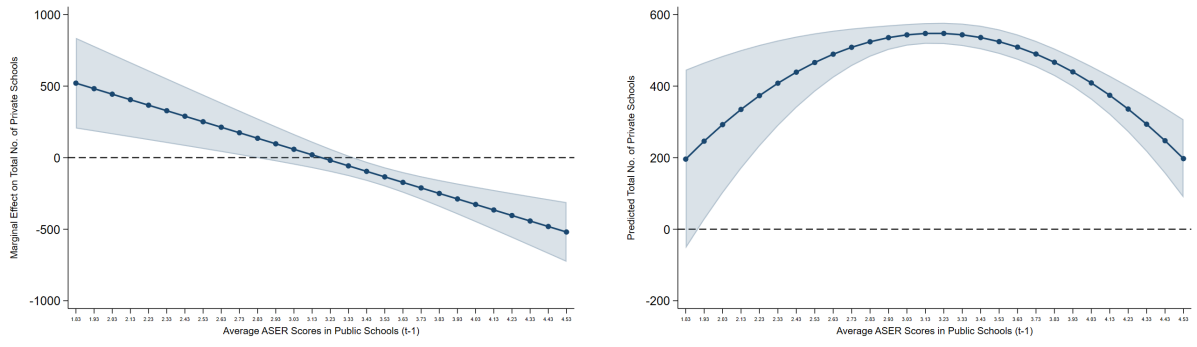


Figure 3: Relationship between lagged public school scores and total no of private schools

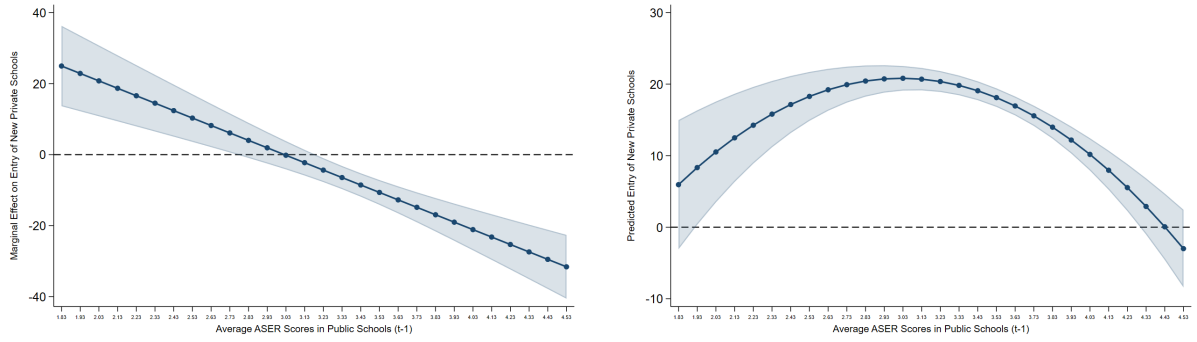


Figure 4: Relationship between lagged public school scores and entry of new private schools

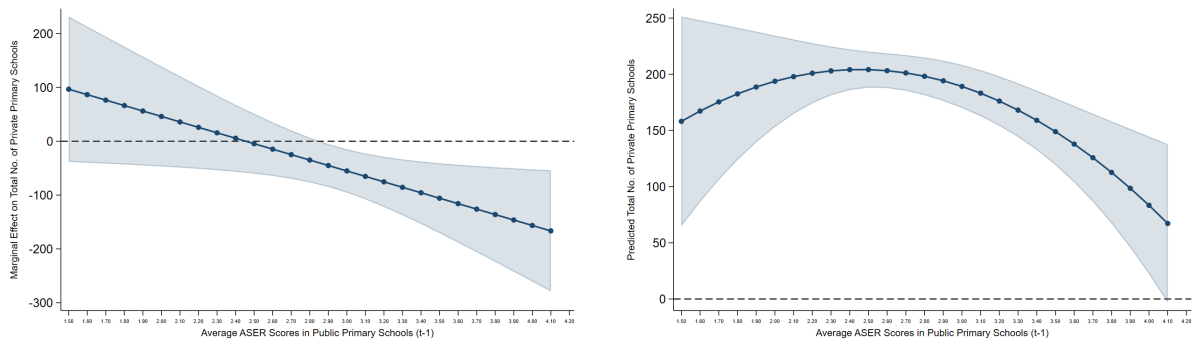


Figure 5: Relationship between lagged public primary school scores and total no of private primary schools

Table 4: Effect of lagged ASER learning levels in public primary schools on no. of private primary schools

	Total No. of Private Primary Schools			
	(1)	(2)	(3)	(4)
Avg ASER learning in public primary schools _{<i>t</i>-1}	-169.8*** (22.62)	-54.19*** (18.40)	311.5* (170.3)	248.1 (162.1)
Avg ASER learning in public primary schools _{<i>t</i>-1} ²			-64.83** (28.74)	-50.62* (27.39)
Average Parental Education _{<i>t</i>-1}		5.399 (9.733)	3.387 (9.452)	2.343 (8.891)
Total no. of public primary schools _{<i>t</i>-1}		0.0205 (0.0193)	0.0272 (0.0194)	0.0285 (0.0191)
Normalized enrollment in public primary schools _{<i>t</i>-1}		-0.167 (0.770)	-0.0415 (0.814)	-0.228 (0.921)
Infrastructure index in public primary schools _{<i>t</i>-1}		22.78 (19.86)	23.56 (19.92)	161.7* (84.17)
Light Intensity _{<i>t</i>-1}		32.70*** (11.93)	32.46*** (11.66)	33.73*** (12.36)
Census infra index		-69.58*** (22.93)	-71.10*** (23.70)	-69.15*** (23.95)
Census infra index × Trend		2.119 (2.117)	2.160 (2.194)	1.518 (2.237)
Log(Rural Population)		76.23*** (27.33)	77.57*** (27.28)	71.22** (28.02)
Log(Rural Population) × Trend		-5.865* (3.080)	-6.458** (3.099)	-6.611** (3.118)
% of SC population		2524.3* (1367.2)	2552.3* (1416.8)	2689.2* (1430.2)
% of SC population × Trend		-234.7 (199.2)	-258.7 (208.7)	-259.8 (208.5)
% of ST population		42.18* (21.74)	36.43 (24.55)	49.26* (27.64)
% of ST population × Trend		8.035 (6.282)	8.750 (6.673)	4.341 (7.025)
Avg ASER learning in public post-primary schools _{<i>t</i>-1}				472.5 (294.2)
Avg ASER learning in public post-primary schools _{<i>t</i>-1} ²				-57.36 (35.53)
Infrastructure index in public post-primary schools _{<i>t</i>-1}				-148.6 (98.74)
Constant	662.3*** (70.21)	-171.0 (207.8)	-640.9* (339.1)	-1444.2** (661.5)
Observations	2457	1958	1958	1958
R-squared	0.104	0.712	0.715	0.720
Mean of Dependent Variable	188.2	188.2	188.2	188.2
# of Blocks & Villages	NO	YES	YES	YES
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Effect of lagged ASER learning levels in public middle schools on total no. of private middle schools

	Total No. of Private Middle Schools			
	(1)	(2)	(3)	(4)
Avg ASER learning in public middle schools _{<i>t</i>-1}	-142.4*** (17.81)	-1.839 (18.76)	579.6*** (190.0)	604.0*** (189.8)
Avg ASER learning in public middle schools _{<i>t</i>-1} ²			-73.58*** (23.09)	-76.40*** (23.16)
Average Parental Education _{<i>t</i>-1}		12.54 (9.022)	11.95 (8.961)	11.61 (8.963)
Total no. of public middle schools _{<i>t</i>-1}		0.109** (0.0525)	0.108** (0.0523)	0.111** (0.0531)
Infrastructure index in public middle schools _{<i>t</i>-1}		33.88* (18.69)	34.38* (18.54)	175.7* (76.69)
Normalized enrollment in public middle schools _{<i>t</i>-1}		-3.155 (3.445)	-3.633 (3.429)	-3.430 (3.408)
Light Intensity _{<i>t</i>-1}		46.63* (24.65)	46.87* (24.47)	47.02* (24.73)
Census infra index		-12.96 (17.67)	-14.96 (17.82)	-15.86 (17.62)
Census infra index × Trend		-0.118 (2.038)	-0.0320 (2.039)	0.0443 (2.023)
Log(Rural Population)		231.7*** (43.23)	232.0*** (42.80)	234.8*** (42.63)
Log(Rural Population) × Trend		-12.02 (7.525)	-12.27 (7.474)	-13.10* (7.417)
% of SC population		6840.6*** (1967.6)	7018.4*** (1968.7)	7165.4*** (1983.0)
% of SC population × Trend		-499.1* (300.7)	-528.9* (303.9)	-559.3* (307.7)
% of ST population		418.0*** (120.3)	423.7*** (122.0)	443.1*** (121.7)
% of ST population × Trend		-16.36 (18.32)	-14.80 (18.74)	-19.26 (18.21)
Avg ASER learning in public secondary schools _{<i>t</i>-1}				-189.7** (93.45)
Avg ASER learning in public secondary schools _{<i>t</i>-1} ²				22.28* (11.72)
Infrastructure index in public secondary schools _{<i>t</i>-1}				-142.0* (74.83)
Constant	764.5*** (75.74)	-1431.7*** (289.0)	-2555.3*** (499.3)	-2190.0*** (511.8)
Observations	2500	1775	1775	1774
R-squared	0.0549	0.700	0.702	0.704
Mean of Dependent Variable	180.9	186.1	186.1	186.2
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
# of Blocks & Villages	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

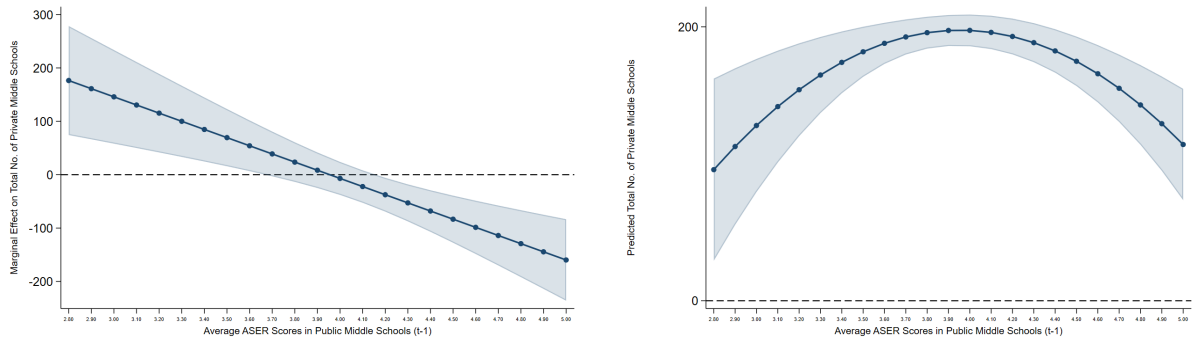


Figure 6: Relationship between lagged public middle school scores and total no of private middle schools

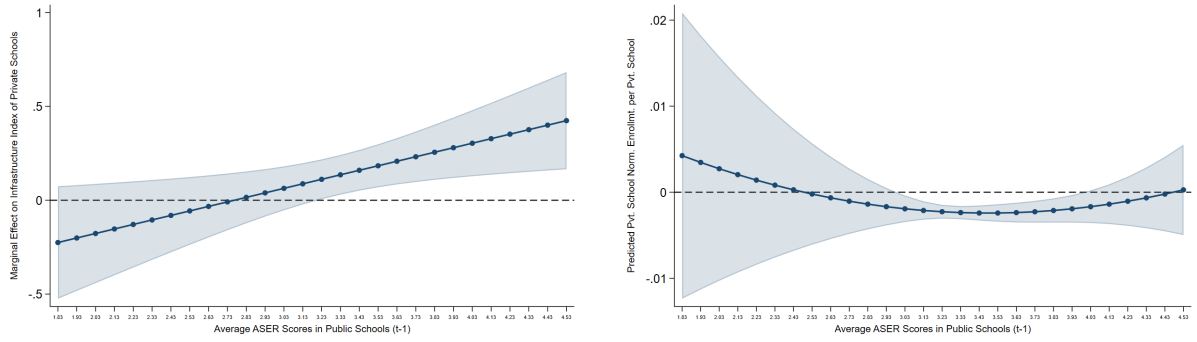


Figure 7: Relationship between lagged public school scores and infrastructure index of private schools

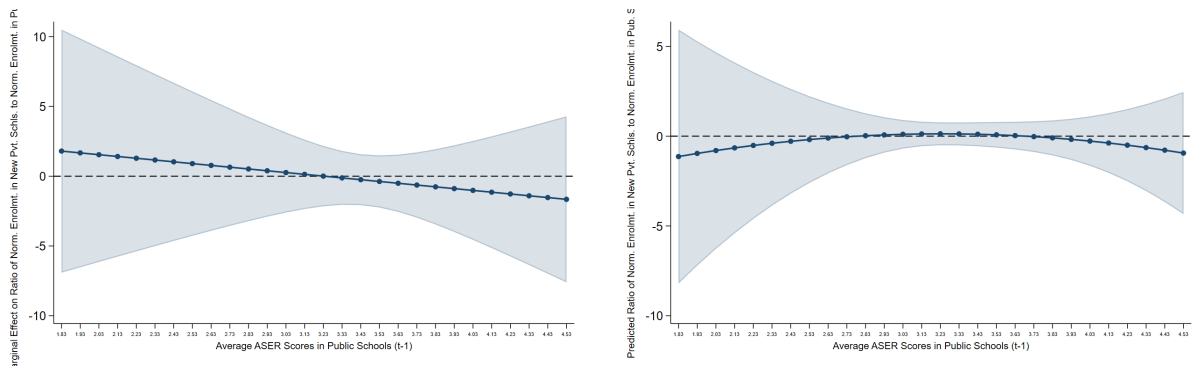


Figure 8: Relationship between lagged public school scores and ratio of enrolment in new private schools to enrolment in public schools

Table 6: Effect of lagged ASER learning levels in public schools on infrastructure index of all private schools

	Private School Infrastructure Index		
	(1)	(2)	(3)
Avg ASER learning in public schools _{t-1}	0.741*** (0.0700)	0.134** (0.0649)	-0.664* (0.388)
Avg ASER learning in public schools _{t-1} ²			0.120** (0.0582)
Average Parental Education _{t-1}		-0.00691 (0.0275)	-0.00435 (0.0273)
Total no. of public schools _{t-1}		-0.0000297 (0.0000459)	-0.0000328 (0.0000460)
Infrastructure index in public schools _{t-1}		0.266*** (0.0604)	0.268*** (0.0602)
Normalized enrollment in public schools _{t-1}		-0.00249 (0.0162)	-0.00255 (0.0163)
Light Intensity _{t-1}		0.276*** (0.0526)	0.286*** (0.0532)
Census infra index		-0.0485 (0.0882)	-0.0444 (0.0882)
Census infra index $\times$ Trend		0.00214 (0.0101)	0.00209 (0.0102)
Log(Rural Population)		0.146 (0.102)	0.143 (0.102)
Log(Rural Population) $\times$ Trend		-0.0175 (0.0112)	-0.0166 (0.0113)
% of SC population		6.598 (8.270)	6.593 (8.294)
% of SC population $\times$ Trend		-0.819 (1.011)	-0.787 (1.030)
% of ST population		-0.635*** (0.118)	-0.625*** (0.121)
% of ST population $\times$ Trend		0.0316 (0.0269)	0.0302 (0.0268)
Constant	0.804*** (0.256)	1.511** (0.724)	2.756*** (0.962)
Observations	2509	1615	1615
R-squared	0.240	0.804	0.805
Mean of Dependent Variable	3.095	3.024	3.024
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE $\times$ Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Effect of lagged ASER learning levels in public schools on ratio of (normalized) enrolment in new private schools to (normalized) enrolment in public schools

	Normalized New Private School Enrollment/Normalized Public School Enrollment		
	(1)	(2)	(3)
Avg ASER learning in public schools _{<i>t-1</i>}	0.0223 (0.786)	-0.130 (1.166)	4.137 (11.07)
Avg ASER learning in public schools _{<i>t-1</i>} ²			-0.640 (1.597)
Average Parental Education _{<i>t-1</i>}		0.410 (0.411)	0.398 (0.411)
Total no. of public schools _{<i>t-1</i>}		0.000454 (0.000775)	0.000485 (0.000771)
Infrastructure index in public schools _{<i>t-1</i>}		1.183 (0.930)	1.162 (0.944)
Light Intensity _{<i>t-1</i>}		-6.751 (7.918)	-6.952 (7.879)
Census infra index		0.679 (1.042)	0.681 (1.044)
Census infra index × Trend		-0.0847 (0.123)	-0.0866 (0.122)
Log(Rural Population)		-3.782*** (1.391)	-3.753*** (1.389)
Log(Rural Population) x Trend		0.337* (0.197)	0.333* (0.195)
% of SC population		-87.64 (75.84)	-88.48 (76.86)
% of SC population × Trend		12.92 (8.921)	12.88 (8.887)
% of ST population		-3.013** (1.412)	-3.072** (1.436)
% of ST population × Trend		0.273 (0.350)	0.286 (0.353)
Constant	0.476 (2.613)	6.849 (7.256)	-0.0114 (19.06)
Observations	1245	978	978
R-squared	0.00360	0.139	0.139
Mean of Dependent Variable	0.0346	-0.00127	-0.00127
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Effect of lagged ASER learning levels in public schools on ratio of (normalized) enrolment in private schools to (normalized) enrolment in public schools

	Normalized Private School Enrollment/Normalized Public School Enrollment		
	(1)	(2)	(3)
Avg ASER learning in public schools _{<i>t</i>-1}	0.0223 (0.786)	-0.130 (1.166)	4.137 (11.07)
Avg ASER learning in public schools _{<i>t</i>-1} ²			-0.640 (1.597)
Average Parental Education _{<i>t</i>-1}		0.410 (0.411)	0.398 (0.411)
Total no. of public schools _{<i>t</i>-1}		0.000454 (0.000775)	0.000485 (0.000771)
Infrastructure index in public schools _{<i>t</i>-1}		1.183 (0.930)	1.162 (0.944)
Light Intensity _{<i>t</i>-1}		-6.751 (7.918)	-6.952 (7.879)
Census infra index		0.679 (1.042)	0.681 (1.044)
Census infra index × Trend		-0.0847 (0.123)	-0.0866 (0.122)
Log(Rural Population)		-3.782*** (1.391)	-3.753*** (1.389)
Log(Rural Population) × Trend		0.337* (0.197)	0.333* (0.195)
% of SC population		-87.64 (75.84)	-88.48 (76.86)
% of SC population × Trend		12.92 (8.921)	12.88 (8.887)
% of ST population		-3.013** (1.412)	-3.072** (1.436)
% of ST population × Trend		0.273 (0.350)	0.286 (0.353)
Constant	0.476 (2.613)	6.849 (7.256)	-0.0114 (19.06)
Observations	1245	978	978
R-squared	0.00360	0.139	0.139
Mean of Dependent Variable	0.0346	-0.00127	-0.00127
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

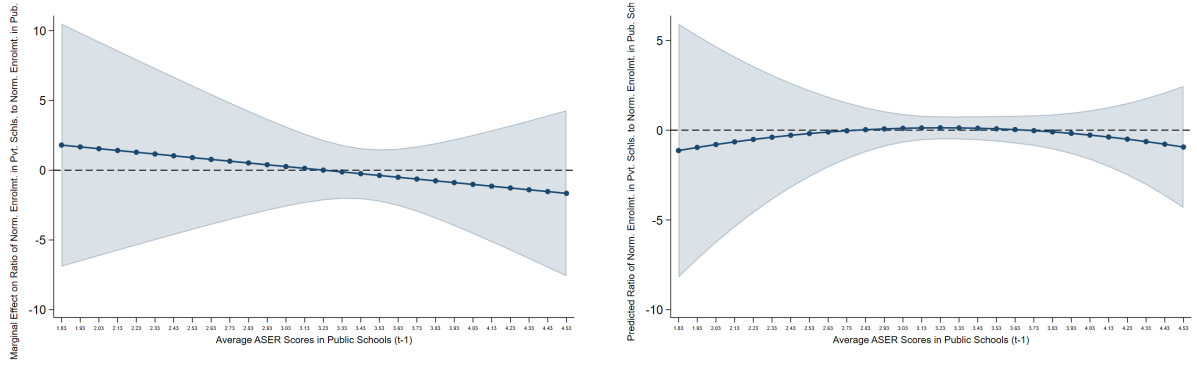


Figure 9: Relationship between lagged public school scores and ratio of enrolment in private schools to enrolment in public schools

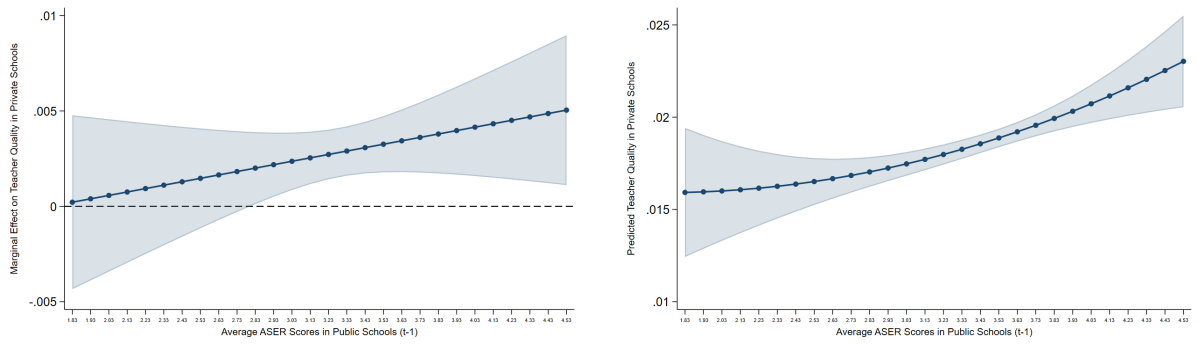


Figure 10: Relationship between lagged public school scores and average teacher quality in private schools

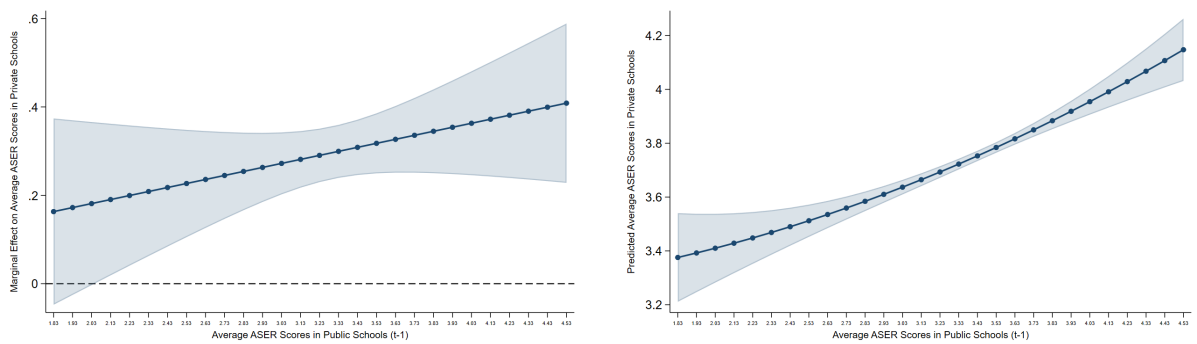


Figure 11: Relationship between lagged public school scores and private school scores

Table 9: Effect of lagged ASER learning levels in public schools on teacher quality in private schools

	Private School Teacher Quality		
	(1)	(2)	(3)
Avg ASER learning in public schools _{<i>t</i>-1}	0.00810*** (0.000793)	0.00289*** (0.000786)	-0.00305 (0.00603)
Avg ASER learning in public schools _{<i>t</i>-1} ²			0.000894 (0.000911)
Average Parental Education _{<i>t</i>-1}		-0.000252 (0.000326)	-0.000233 (0.000324)
Total no. of public schools _{<i>t</i>-1}		-0.000000391 (0.000000659)	-0.000000414 (0.000000661)
Infrastructure index in public schools _{<i>t</i>-1}		0.00214*** (0.000691)	0.00215*** (0.000691)
Normalized enrollment in public schools _{<i>t</i>-1}		-0.000115 (0.000189)	-0.000116 (0.000191)
Light Intensity _{<i>t</i>-1}		0.00451** (0.00176)	0.00458*** (0.00176)
Census infra index		0.0000121 (0.000851)	0.0000422 (0.000847)
Census infra index × Trend		0.000141 (0.0000902)	0.000141 (0.0000901)
Log(Rural Population)		0.00305** (0.00135)	0.00303** (0.00135)
Log(Rural Population) × Trend		-0.000225 (0.000143)	-0.000218 (0.000143)
% of SC population		-0.113 (0.0965)	-0.113 (0.0972)
% of SC population × Trend		-0.00998 (0.0138)	-0.00975 (0.0140)
% of ST population		-0.00119 (0.00117)	-0.00112 (0.00120)
% of ST population × Trend		0.0000251 (0.000303)	0.0000146 (0.000311)
Constant	-0.00894*** (0.00257)	-0.0101 (0.00922)	-0.000891 (0.0136)
Observations	2514	1615	1615
R-squared	0.123	0.721	0.721
Mean of Dependent Variable	0.0188	0.0186	0.0186
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE × Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Effect of lagged ASER learning levels in public schools on average learning in private schools

	(1)	(2)	(3)
Avg ASER learning in public schools _{t-1}	0.253*** (0.0188)	0.299*** (0.0368)	-0.00269 (0.278)
Avg ASER learning in public schools _{t-1} ²			0.0454 (0.0419)
Average Parental Education _{t-1}		0.0303* (0.0166)	0.0318* (0.0166)
Total no. of public schools _{t-1}		-0.00000210 (0.0000252)	-0.00000163 (0.0000252)
Infrastructure index in public schools _{t-1}		0.0249 (0.0303)	0.0257 (0.0307)
Normalized enrollment in public schools _{t-1}		0.00239 (0.0106)	0.00273 (0.0106)
Light Intensity _{t-1}		0.102 (0.0652)	0.106 (0.0658)
Census infra index		0.0270 (0.122)	0.0228 (0.122)
Census infra index $\times$ Trend		-0.0255 (0.0331)	-0.0237 (0.0333)
Log(Rural Population)		-0.0314 (0.150)	-0.0355 (0.150)
Log(Rural Population) $\times$ Trend		-0.00443 (0.0414)	-0.00304 (0.0413)
% of SC population		28.55*** (10.81)	28.50*** (10.89)
% of SC population $\times$ Trend		-7.230** (2.878)	-7.221** (2.900)
% of ST population		-1.771** (0.731)	-1.838** (0.735)
% of ST population $\times$ Trend		0.431** (0.182)	0.452** (0.184)
Constant	2.871*** (0.0645)	2.858*** (0.421)	3.321*** (0.592)
Observations	1460	760	760
R-squared	0.141	0.480	0.481
Mean of Dependent Variable	3.744	3.744	3.744
Year FE	NO	YES	YES
State FE	NO	YES	YES
# of Blocks & Villages	NO	YES	YES
State FE $\times$ Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Effect of lagged ASER learning levels in govt primary schools on entry of private primary schools

	Private Primary School Entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in public primary schools _{t-1}	-5.005*** (0.985)	-4.018*** (1.369)	18.93** (8.768)	16.48* (9.835)
Avg ASER learning in public primary schools _{t-1} ²			-4.069*** (1.486)	-3.346** (1.691)
Average Parental Education _{t-1}		-0.00928 (0.631)	-0.136 (0.616)	-0.226 (0.601)
Total no. of public primary schools _{t-1}		0.00430*** (0.00155)	0.00471*** (0.00153)	0.00488*** (0.00153)
Normalized enrollment in public primary schools _{t-1}		0.117*** (0.0363)	0.125*** (0.0308)	0.100*** (0.0255)
Infrastructure index in public primary schools _{t-1}		0.484 (1.058)	0.531 (1.065)	14.36*** (2.961)
Light Intensity _{t-1}		4.387*** (0.648)	4.374*** (0.637)	4.490*** (0.643)
Census infra index		-3.070 (2.226)	-3.184 (2.227)	-2.965 (2.233)
Census infra index × Trend		0.245 (0.334)	0.249 (0.342)	0.185 (0.350)
Log(Rural Population)		13.14*** (3.336)	13.22*** (3.333)	12.60*** (3.355)
Log(Rural Population) × Trend		-1.949*** (0.578)	-1.988*** (0.581)	-1.997*** (0.583)
% of SC population		394.1** (161.6)	396.3** (162.9)	407.9** (162.5)
% of SC population × Trend		-55.59** (28.23)	-57.24** (28.54)	-57.05** (28.60)
% of ST population		13.43*** (4.588)	13.07*** (4.642)	14.33*** (5.147)
% of ST population × Trend		-2.076** (0.884)	-2.035** (0.903)	-2.467** (1.029)
Avg ASER learning in public post-primary schools _{t-1}				30.93 (20.30)
Avg ASER learning in public post-primary schools _{t-1} ²				-3.749 (2.476)
Infrastructure index in public post-primary schools _{t-1}				-14.87*** (3.064)
Constant	22.75*** (2.918)	-11.97 (13.65)	-41.29** (20.83)	-94.23** (41.39)
Observations	1979	1965	1965	1965
R-squared	0.0248	0.514	0.517	0.529
Mean of Dependent Variable	8.935	8.980	8.980	8.980
# of Blocks & Villages	NO	YES	YES	YES
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

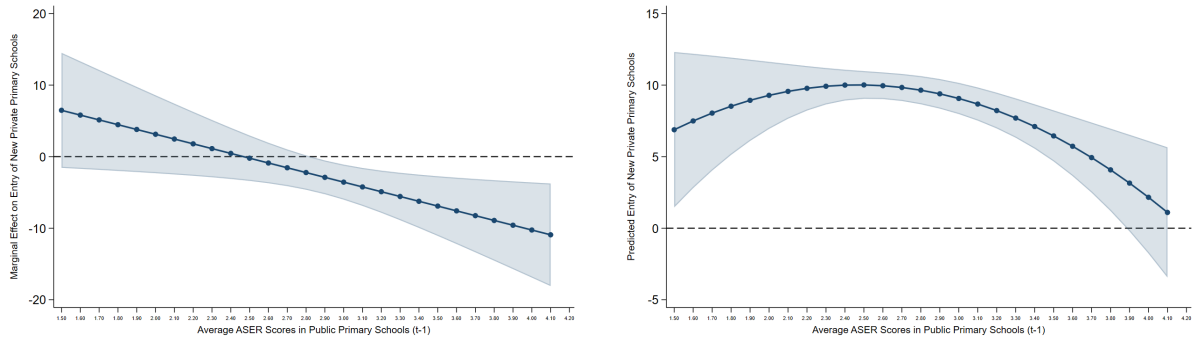


Figure 12: Relationship between lagged public primary school scores and entry of private primary schools

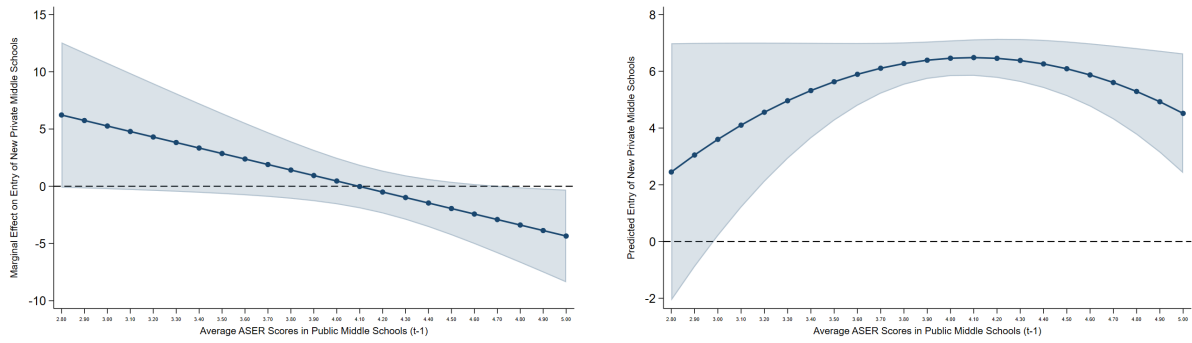


Figure 13: Relationship between lagged public middle school scores and entry of private middle schools

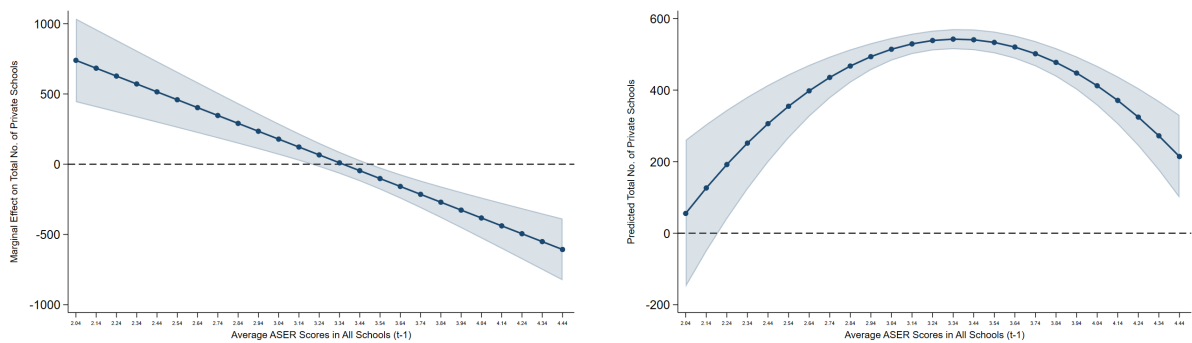


Figure 14: Relationship between lagged scores in all schools and total no. of private schools

Table 12: Effect of lagged ASER learning levels in public middle schools on entry of new private middle schools

	Private Middle School Entry			
	(1)	(2)	(3)	(4)
Avg ASER learning in public middle schools _{<i>t</i>-1}	-6.114*** (1.207)	-0.335 (1.152)	21.56* (11.23)	19.67* (11.22)
Avg ASER learning in public middle schools _{<i>t</i>-1} ²			-2.772** (1.340)	-2.403* (1.335)
Average Parental Education _{<i>t</i>-1}		0.671 (0.530)	0.648 (0.532)	0.607 (0.536)
Total no. of public middle schools _{<i>t</i>-1}		0.00362* (0.00202)	0.00360* (0.00201)	0.00370* (0.00201)
Infrastructure index in public middle schools _{<i>t</i>-1}		0.101 (1.059)	0.121 (1.056)	11.93** (4.674)
Normalized enrollment in public middle schools _{<i>t</i>-1}		0.146 (0.190)	0.127 (0.188)	0.138 (0.190)
Light Intensity _{<i>t</i>-1}		1.290* (0.737)	1.300* (0.729)	1.310* (0.754)
Census infra index		-1.229 (1.320)	-1.298 (1.325)	-1.232 (1.328)
Census infra index × Trend		0.235 (0.198)	0.236 (0.199)	0.202 (0.202)
Log(Rural Population)		7.494*** (2.639)	7.503*** (2.633)	7.512*** (2.599)
Log(Rural Population) × Trend		-0.0118 (0.438)	-0.0215 (0.437)	-0.0620 (0.429)
% of SC population		193.3 (121.5)	198.9 (121.2)	197.4 (120.2)
% of SC population × Trend		-0.476 (19.49)	-1.316 (19.47)	-1.381 (19.29)
% of ST population		11.70* (6.333)	11.94* (6.359)	13.50** (6.392)
% of ST population × Trend		0.408 (1.028)	0.460 (1.034)	0.0315 (1.017)
Avg ASER learning in public secondary schools _{<i>t</i>-1}				2.082 (5.791)
Avg ASER learning in public secondary schools _{<i>t</i>-1} ²				-0.374 (0.739)
Infrastructure index in public secondary schools _{<i>t</i>-1}				-11.94*** (4.460)
Constant	30.83*** (5.175)	-58.06*** (15.35)	-100.3*** (26.88)	-98.06*** (29.41)
Observations	2008	1777	1777	1776
R-squared	0.0391	0.473	0.475	0.478
Mean of Dependent Variable	6.050	6.134	6.134	6.137
Year FE	NO	YES	YES	YES
State FE	NO	YES	YES	YES
# of Blocks & Villages	NO	YES	YES	YES
State FE × Year FE	NO	YES	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Effect of lagged ASER learning levels in all schools on total no. of private schools

	Total Private Schools		
	(1)	(2)	(3)
Avg ASER learning in all schools _{t-1}	-157.6*** (54.71)	11.14 (49.97)	1883.0*** (432.5)
Avg ASER learning in public schools _{t-1} ²			-280.5*** (62.61)
Average Parental Education _{t-1}		41.43** (19.17)	33.81* (18.52)
Total no. of public schools _{t-1}		-0.0224 (0.0341)	-0.0106 (0.0332)
Normalized enrollment in public schools _{t-1}		-6.159 (10.24)	-7.338 (10.23)
Infrastructure index in public schools _{t-1}		14.35 (44.84)	10.65 (43.83)
Light Intensity _{t-1}		319.6** (131.6)	304.0** (129.3)
Census infra index		-156.0*** (48.91)	-148.2*** (48.09)
Census infra index $\times$ Trend		15.15*** (5.512)	13.34** (5.545)
Log(Rural Population)		383.7*** (72.66)	395.8*** (70.74)
Log(Rural Population) $\times$ Trend		-12.76 (9.946)	-16.11 (9.847)
% of SC population		11854.3*** (3557.0)	12290.2*** (3486.6)
% of SC population $\times$ Trend		-718.5 (525.0)	-902.1* (531.0)
% of ST population		214.2*** (67.83)	190.8*** (64.00)
% of ST population $\times$ Trend		29.23 (22.38)	31.94 (21.87)
Constant	1069.8*** (192.2)	-2465.1*** (549.6)	-5427.3*** (927.2)
Observations	2514	1615	1615
R-squared	0.00984	0.753	0.760
Mean of Dependent Variable	528.6	501.8	501.8
# of Blocks & Villages	NO	YES	YES
Year FE	NO	YES	YES
State FE	NO	YES	YES
State FE $\times$ Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Effect of lagged average ASER learning levels in all schools on the entry of private schools

	Private School Entry		
	(1)	(2)	(3)
Avg ASER learning _{t-1}	-9.614*** (2.024)	-6.164** (2.472)	83.92*** (19.82)
Avg ASER learning _{t-1} ²			-13.50*** (2.948)
Average Parental Education _{t-1}		0.607 (0.952)	0.240 (0.914)
Total no. of govt schools _{t-1}		0.00689*** (0.00212)	0.00746*** (0.00210)
Normalized enrollment in govt schools _{t-1}		-0.0751 (0.734)	-0.132 (0.727)
Infrastructure index in govt schools _{t-1}		-0.598 (1.920)	-0.776 (1.871)
Light Intensity _{t-1}		8.772*** (2.605)	8.017*** (2.607)
Census infra index		-7.048** (3.247)	-6.673** (3.173)
Census infra index $\times$ Trend		0.847 (0.525)	0.760 (0.526)
log(Rural population)		17.56*** (4.123)	18.14*** (4.083)
log(Rural population) $\times$ Trend		-1.448** (0.609)	-1.610*** (0.610)
% of SC population		530.1** (208.4)	551.0*** (204.0)
% of SC population $\times$ Trend		-34.31 (31.29)	-43.14 (31.55)
% of ST population		13.02*** (4.153)	11.89*** (4.099)
% of ST population $\times$ Trend		-0.590 (0.880)	-0.460 (0.872)
Constant	51.07*** (7.092)	-49.33** (21.67)	-191.9*** (40.29)
Observations	2019	1615	1615
R-squared	0.0202	0.499	0.508
Mean of Dependent Variable	18.30	17.39	17.39
# of Blocks & Villages	NO	YES	YES
Year FE	NO	YES	YES
State FE	NO	YES	YES
State FE $\times$ Year FE	NO	YES	YES

Standard errors clustered at district level in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

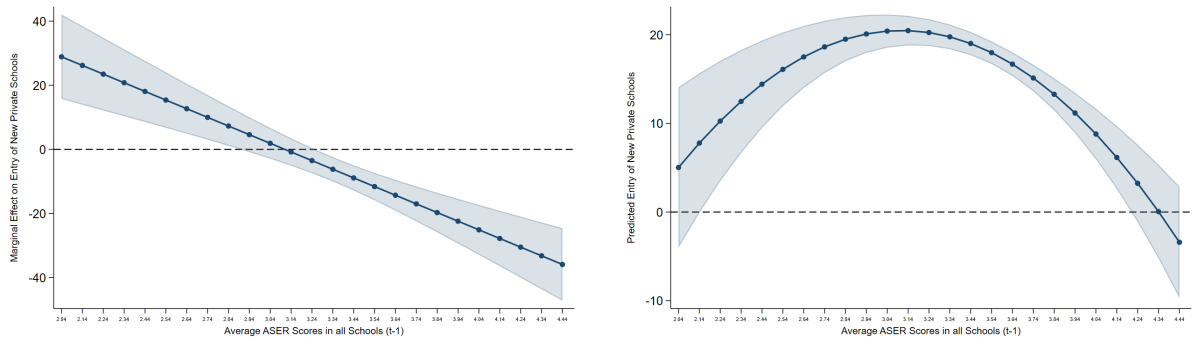


Figure 15: Relationship between lagged scores in all schools and entry of new private schools